

Method of Undetermined Coefficients:

↳ For: 1st or higher order ODEs + Non-homogeneous
+ Linear

↳ The general solution for these DEs have two parts:

$$y = y_c + y_p$$

general solution ← ↓ complementary solution → particular integral

↳ The complementary solution, y_c , is first obtained by solving the homogeneous version of the given DE. This is obtained by setting the non-homogeneous part to zero i.e. $g(x)=0$.

* Method of solving the homogeneous version is the one we have seen in the previous lecture.

↳ The non-homogeneous part, $g(x)$, of this DE can be a constant, a polynomial function, an exponential function, sine or cosine function or even finite sums and products of these functions.

↳ Based on the form of the non-homogeneous part, $g(x)$, the form of the particular solution, y_p , will vary.

* The particular solution is also called the particular integral.

→ The form of the particular solution can be determined using the following table:

Form of $g(x)$ (RHS)	Form of particular sol ⁿ ^{trial}
$g(x) = a_0$ (just a constant)	$y_p = A$
$g(x) = a_1 x + a_0$ (linear polynomial)	$y_p = Ax + B$
$g(x) = a_2 x^2 + a_1 x + a_0$ (quadratic polynomial)	$y_p = Ax^2 + Bx + C$
$g(x) = a_3 x^3 + a_2 x^2 + a_1 x + a_0$ (cubic)	$y_p = Ax^3 + Bx^2 + Cx + D$
$g(x) = a_0 \cdot e^{\lambda x}$	$y_p = A \cdot e^{\lambda x}$ ← note that the power doesn't change
$g(x) = a_1 \cos(\lambda x) + a_0 \sin(\lambda x)$	$y_p = A \cos(\lambda x) + B \sin(\lambda x)$ note that the argument inside doesn't change
$g(x) = a_0 \cos(\lambda x)$	$y_p = A \cos(\lambda x) + B \sin(\lambda x)$
$g(x) = a_0 \sin(\lambda x)$	$y_p = A \cos(\lambda x) + B \sin(\lambda x)$
$g(x) = \underline{a_1 \cos(\lambda x)} + \underline{a_0 \sin(\mu x)}$ different	$y_p = \underline{A \cos(\lambda x) + B \sin(\lambda x)}$ + $\underline{C \cos(\mu x) + D \sin(\mu x)}$ this line for $a_1 \cos \lambda x$ this line for $a_0 \sin \mu x$
$g(x) = \underline{a_3 e^{\lambda x}} + \underline{a_2 x^2 + a_1 x + a_0}$	$y_p = \underline{A e^{\lambda x}} + \underline{B x^2 + C x + D}$
$g(x) = a_0 \cdot \underline{e^{\lambda x}} \cdot \underline{\cos(\mu x)}$	$y_p = \underline{e^{\lambda x}} (\underline{A \cos(\mu x) + B \sin(\mu x)})$
$g(x) = a_1 \underline{e^{\lambda x}} + a_0 \underline{e^{\mu x}}$	$y_p = \underline{A e^{\lambda x}} + \underline{B e^{\mu x}}$
$g(x) = a_0 \cdot \underline{e^{\lambda x}} \cdot \underline{e^{\mu x}}$ = $a_0 \underline{e^{(\lambda+\mu)x}}$	$y_p = A \underline{e^{(\lambda+\mu)x}}$

* The non-homogeneous part may also be sums of the possible for as seen from the 9th case onwards. If it is so, then y_p will also be sums of their respective particular solⁿ form.

↳ We take the appropriate form of the particular solⁿ as the trial solution and substitute it into the non-homogeneous version of the DE and then solve to find the coefficients $\lambda / \lambda_i / \lambda, \mu$

(e.g.) Solve: $y'' + 4y' - 2y = 2x^2 - 3x + 6$

Solⁿ: The general solution is given by:

$$y = y_c + y_p$$

→ The complementary solution, y_c , is obtained by

solving: $y'' + 4y' - 2y = 0$

Let $y = e^{mx}$ be a trial solution of the homogeneous equation. Substituting this into the DE:

$$m^2 \cdot e^{mx} + 4m \cdot e^{mx} - 2e^{mx} = 0$$

$$\Rightarrow m^2 + 4m - 2 = 0$$

$$\Rightarrow m = \frac{-4 \pm \sqrt{4^2 - 4 \cdot 1 \cdot (-2)}}{2 \cdot 1} = \frac{-4 \pm \sqrt{16 + 8}}{2}$$

$$= \frac{-4 \pm 2\sqrt{6}}{2} = -2 \pm \sqrt{6}$$

$$\therefore y_c = C_1 \cdot e^{(-2+\sqrt{6})x} + C_2 \cdot e^{(-2-\sqrt{6})x}$$

→ Since the non-homogeneous part is a quadratic equation, the particular solution is of the form:

$$y_p = Ax^2 + Bx + C$$

$$\text{Now, } y'_p = 2Ax + B$$

$$y''_p = 2A$$

Substituting y_p into the DE, we get:

$$y'' + 4y' - 2y = 2x^2 - 3x + 6$$

$$\Rightarrow (2A) + 4(2Ax + B) - 2(Ax^2 + Bx + C) = 2x^2 - 3x + 6$$

$$\Rightarrow 2A + 8Ax + 4B - 2Ax^2 - 2Bx - 2C = 2x^2 - 3x + 6$$

$$\Rightarrow (-2A)x^2 + (8A - 2B)x + (2A + 4B - 2C) = 2x^2 - 3x + 6$$

$\therefore$ Equating the coefficients:

$$-2A = 2$$

$$8A - 2B = -3$$

$$2A + 4B - 2C = 6$$

Solving the above system: $A = -1$, $B = -\frac{5}{2}$, $C = -9$

$$\therefore y_p = -x^2 - \frac{5}{2}x - 9$$

$\hookrightarrow$ Hence, the general solution is:

$$y = y_c + y_p$$

$$= C_1 \cdot e^{(-2+\sqrt{6})x} + C_2 \cdot e^{(-2-\sqrt{6})x} - x^2 - \frac{5}{2}x - 9$$

Ex. 9 Solve $y'' + y = 4x + 10 \sin x$, given $y(\pi) = 0$, $y'(\pi) = 2$.

Soln $\rightarrow$ To find y_c , we solve: $y'' + y = 0$

Let $y = e^{mx}$. Then the homogeneous DE becomes:

$$m^2 + 1 = 0$$

$$\Rightarrow m = \pm \sqrt{-1} = \pm i = 0 \pm 1i$$

The trial solution hence becomes:

$$y_c = e^{0x} \cdot (C_1 \cdot \cos(1x) + C_2 \cdot \sin(1x))$$

$$= C_1 \cdot \cos x + C_2 \sin x$$

$\rightarrow$ For the particular solution, according to the non-homogeneous part of the DE:

$$y_p = Ax + B + C \cdot \cos x + D \cdot \sin x$$

$$y_p' = A - C \cdot \sin x + D \cos x$$

$$y_p'' = -C \cdot \cos x - D \cdot \sin x$$

Substituting y_p into the original DE:

$$y_p'' + y_p = 4x + 10 \sin x$$

$$\Rightarrow (-C \cdot \cos x - D \cdot \sin x) + (Ax + B + C \cdot \cos x + D \cdot \sin x)$$

$$= 4x + 10 \sin x$$

$$\Rightarrow Ax + B + (-D + D) \sin x + (-C + C) \cos x = 4x + 10 \sin x$$

$$\Rightarrow Ax + B = 4x + 10 \sin x \quad \leftarrow \text{This can't be solved}$$

↳ This happens because of the repetition of $\cos x$ and $\sin x$ in y_p .

$$\therefore \text{Let } y_p = Ax + B + x(C \cdot \cos x + D \cdot \sin x)$$

$$= Ax + B + Cx \cdot \cos x + D \cdot x \cdot \sin x$$

$$y_p' = A + C \cdot \cos x - C \cdot x \cdot \sin x + D \cdot \sin x + D \cdot x \cdot \cos x$$

$$= A + (C + Dx) \cos x + (D - Cx) \sin x$$

$$y_p'' = D \cos x - (C + Dx) \sin x + (-C) \sin x + (D - Cx) \cos x$$

$$= (2D - Cx) \cos x + (-2C - Dx) \sin x$$

Substituting y_p into the DE:

$$y_p'' + y_p = 4x + 10 \sin x$$

$$\Rightarrow (2D - Cx) \cos x + (-2C - Dx) \sin x + Ax + B$$

$$+ Cx \cdot \cos x + Dx \cdot \sin x = 4x + 10 \sin x$$

$$\Rightarrow 2D \cdot \cos x - 2C \cdot \sin x + Ax + B = 4x + 10 \sin x$$

$\therefore$ Equating the coefficients:

$$A = 4$$

$$B = 0$$

$$-2C = 10 \Rightarrow C = -5$$

$$2D = 0 \Rightarrow D = 0$$

$$\text{Hence, } y_p = 4x - 5x \cos x$$

$$\text{and } y = y_c + y_p$$

$$= C_1 \cdot \cos x + C_2 \sin x + 4x - 5x \cos x$$

is the general solution.

→ Now, given initial conditions:

$$y(\pi) = 0$$

$$\Rightarrow C_1 \cdot \cos(\pi) + C_2 \cdot \sin(\pi) + 4\pi - 5\pi \cos(\pi) = 0$$

$$\Rightarrow -C_1 + C_2 \cdot 0 + 4\pi + 5\pi = 0$$

$$\Rightarrow C_1 = 9\pi$$

$$\text{Then, } y'(\pi) = 2$$

$$\text{So } y = C_1 \cdot \cos x + C_2 \sin x + 4x - 5x \cos x$$

$$\begin{aligned} \Rightarrow y' &= -C_1 \cdot \sin x + C_2 \cdot \cos x + 4 - 5 \cos x + 5x \sin x \\ &= (-C_1 + 5x) \cdot \sin x + (C_2 - 5) \cos x + 4 \end{aligned}$$

$$\therefore y'(\pi) = (-C_1 + 5\pi) \sin(\pi) + (C_2 - 5) \cos \pi + 4$$

$$\Rightarrow 2 = 0 - (C_2 - 5) + 4$$

$$\Rightarrow C_2 - 5 = 2 \quad \Rightarrow C_2 = 7$$

Hence, the solution is:

$$y = 9\pi \cdot \cos x + 7 \cdot \sin x + 4x - 5x \cos x$$

